SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
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Section / Batch:		Date:	

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
		GRAND TOTAL:	100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.
(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.
(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I ________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding major misconceptions
Explanation	25%	Detailed well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks							
					Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

(a) City Hospital Data warehouse

Snowflake Schema

Patient - Dim
↓
Doctor - Specialization - Doctor - Dim - Department - Dim - hospital
↓
Fact - Visit
↓
Patient - ID
↓
Doctor - ID
↓
Department - ID
↓
Time - ID
↓
Visit - ID
↓
Avg - ID
↓
Time - Dim
↓
Day - Month - Quarter - year

(b) OLAP operations :-

* Roll up Time ⟶ year
* Roll up Patient ⟶ All
* Roll up Doctor ⟶ All
* keep Department dimension unchanged

* Aggregate Avg - charge

output:

|Department|year| Average charge|

(e) Number of cuboids:-

Formula:- Total cuboids = product of hierarchy levels.

Each dimensions = 4

$4^4 = 256$

(2) Smart city star schema:-

Star Schema:-

Time - Dim

↓

Traffic - Dim — Smart city - fact — Location - Dim

↓

Transport - Dim

↓

citizen - Dim.

Fact table:-

Measures	Time	Location	Traffic	Transport
vehicle count	year	city	Sensor ID	Bus
congestion level	month	zone	Road type	metro
Energy consumption.	Day	area.		

Air quality index hour
Noise level
Ridership
Punctuality

Justification:-

* measures store numerical values
* Dimensions enable multidimensional analysis by Location, time, transport and traffic.

③ Banking Data warehouse:-

④ Galaxy Schema:-

Customer-Dim
↓
Branch Dim – Transaction-fact – Time-Dim
↓
Account type-Dim
↓
Customer Dim
↓
Branch Dim – Loan-fact – Time-Dim.
↓
Account-type-Dim

Transaction Fact

Transaction count

Amount

Loan Fact

Loan amount

Interest

ⓑ OLAP operation:

Starting cuboid: Customer, Branch, Time, Account type

Operations:-

* Roll up Time → year
* Roll up customer → All
* Roll up Account type → All
* Keep Branch
* Aggregate Average Amount and loan amount

Output:-

| Branch | Year | Avg transaction | Avg loan |

ⓒ cuboids

4 dimensions

each has 4 levels

$4^4 = 256$

(1) Manufacturing company :-

Roll-up

Roll-up Product :- Batch → type → category

Roll-up factory :- line → Department → category

keep current year.

Output :- Average defect rate by product category.

Drill-down factory

Factory
↓
Department

Product

category
↓
type

slide :-

Condenzation :- Supplier Rating ≥ 4

Display :- production unit.

Dice :-

Select

* product categories
* factory Location

measures

* Defect Rate
* Supplier reliability %

(5) Retail chain:-

Roll-up:-

Roll-up:- SKU → Subcategory → category

Time:- Day → Month → Quaters

measure:- Inventory Turnover

Drill-down

Region
↓
city
↓
Store

Electronics
↓
Sub-category
↓
SKU

slice:-

Condition:-

Gender = male

Membership = Bronze

Measure:-

Sales Revenue

Dice:-

Selected

* cities
* Product categories
* Time Period

Measure

* Sales Revenge
* Units Solid.